



Important Formulas: Perimeter and Area

Table of Contents

1. Perimeter
2. Perimeter of Common Plane Figures
3. Area
4. Summary

Perimeter

The perimeter of a figure is the total length of its boundary.

It is the distance around a closed shape.

Formula (General):

$$\text{Perimeter} = \text{Sum of all sides}$$

Example:

If a park has boundary lengths 100 m, 75 m, 200 m, 220 m, and 80 m:

$$\text{Perimeter} = 100 + 75 + 200 + 220 + 80 = 675 \text{ m}$$

NOTE: The word Perimeter comes from Greek -
'Peri' means "around" and 'Metron' means "measure".

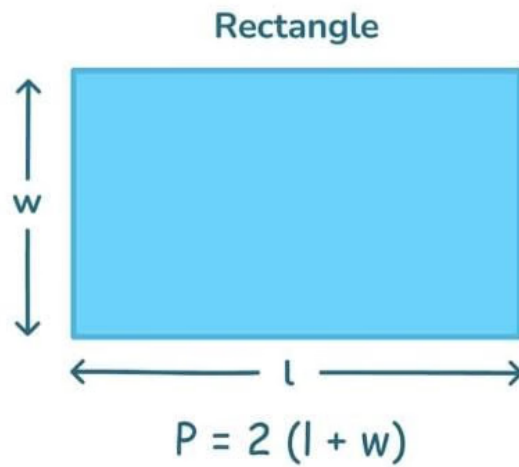
Perimeter of Common Plane Figures

1. Perimeter of a Rectangle:

Add the length and width.

Multiply the sum by 2.

Formula: Perimeter = $2 \times (L + W)$



Example: Perimeter of rectangle ABCD = $2(l + b)$

If length = 14 cm and perimeter = 52 cm,

then $52 = 2(14 + b)$

$26 = 14 + b$

$b = 12$ cm

Breadth of the rectangle is 12 cm.

2. Perimeter of a Square:

All sides are equal.

Formula: Perimeter = $4 \times \text{side}$



Example: If each side of a square = 7 cm,

$$\text{Perimeter} = 4 \times 7 = 28 \text{ cm}$$

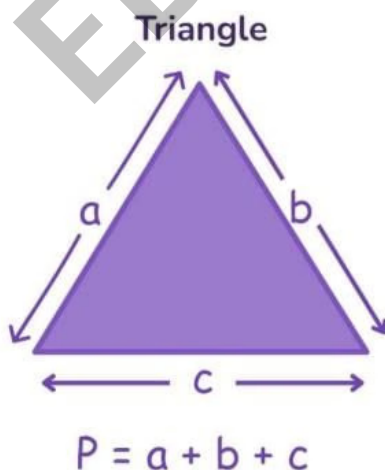
Another Example: If the perimeter of a square = 160 cm,

$$\text{side} = 160 \div 4 = 40 \text{ cm}$$

3. Perimeter of a Triangle:

Add all the sides.

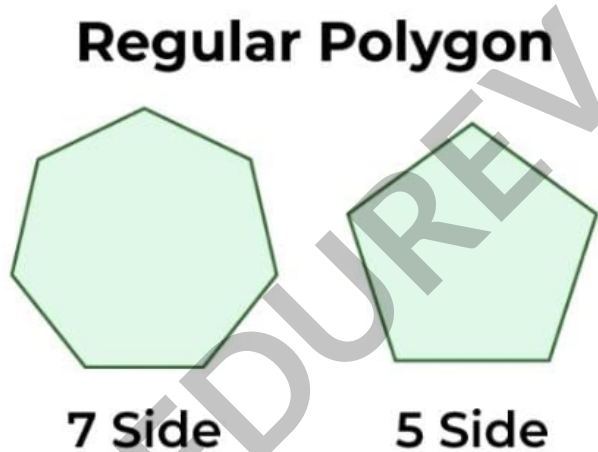
Formula: Perimeter = $a + b + c$



Example: A triangle with sides 4 cm, 5 cm, and 7 cm has a perimeter of 16 cm.

4. Perimeter of a Regular Polygon:

Multiply the length of one side by the number of sides.



Formula: Perimeter = number of sides $\times$ side length

Example: For a regular hexagon with 6 sides of 5 cm each,

$$\text{Perimeter} = 6 \times 5 = 30 \text{ cm}$$

Area

The area is the space inside a shape.

It is measured in square units (like cm^2 or m^2).

Example:

The number of square tiles needed to cover a floor gives its area.

General Formula (using unit squares):

Area = Number of square units covering the surface

5. Area of a Square:

Multiply side by side.

Formula: Area = side $\times$ side



Square

$$A = a^2$$

Example:

If each side of a square = 5 cm,

Area = $5 \times 5 = 25$ sq cm

Another Example:

If each side of a square field = 70 m,

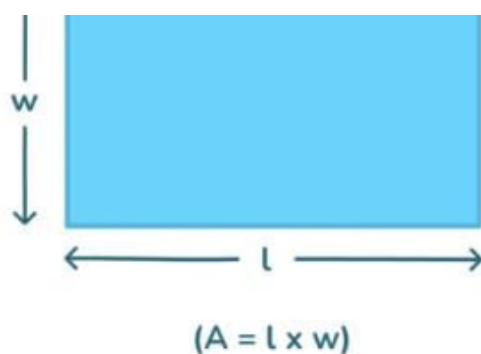
Area = $70 \times 70 = 4900$ sq m

6. Area of a Rectangle:

Multiply length by width.

Formula: Area = $L \times W$



**Example:**

If length = 12 cm and breadth = 4 cm,

Area = $12 \times 4 = 48$ sq cm

Another Example:

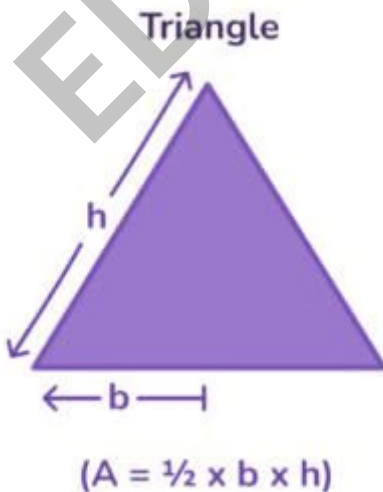
If area = 48 sq cm and breadth = 8 cm,

Length = Area $\div$ Breadth = $48 \div 8 = 6$ cm

7. Area of a Triangle:

The area of a triangle is half of a rectangle.

Formula: Area = $\frac{1}{2} \times \text{base} \times \text{height}$



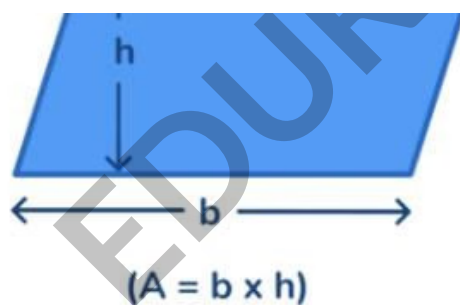
Example: If base = 6 cm and height = 4 cm, area = $\frac{1}{2} \times 6 \times 4 = 12$ cm²

8. Area of a Parallelogram

Area = base $\times$ height

Parallelogram



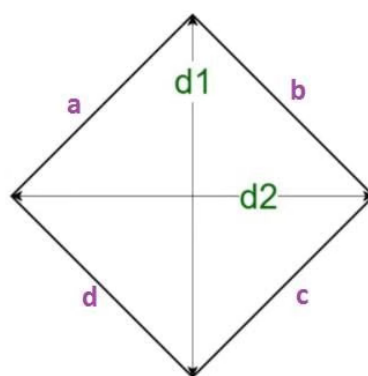
**Example:**

If the base of a parallelogram = 9 cm and height = 6 cm,

$$\text{Area} = 9 \times 6 = 54 \text{ sq cm}$$

9. Area of a Rhombus

Area = $\frac{1}{2} \times$ (product of diagonals)



$$\text{Area} = \frac{d_1 \times d_2}{2}$$

Example:

If the diagonals of a rhombus are 10 cm and 8 cm,

$$\text{Area} = \frac{1}{2} \times 10 \times 8 = 40 \text{ sq cm}$$

Summary

Shape	Perimeter Formula	Area Formula
Triangle	$P = a + b + c$	$A = \frac{1}{2} \times \text{base} \times \text{height}$
Rectangle	$P = 2(l + b)$	$A = l \times b$
Square	$P = 4 \times \text{side}$	$A = \text{side} \times \text{side}$

Regular Polygon	$P = n \times \text{side}$	–
Parallelogram	–	$A = \text{base} \times \text{height}$
Rhombus	–	$A = \frac{1}{2} \times d_1 \times d_2$